

Assignment 1: Core Python Programming

1. Write a Python program to declare variables of different data types, display their values, and print their data types.
2. Write a Python program to accept two numbers from the user, perform all arithmetic operations, and display the results.
3. Write a Python program to compare two numbers, display the comparison results, and identify the larger number.
4. Write a Python program to print numbers from 1 to 50, display all even numbers, and calculate their sum.
5. Write a Python program to generate the multiplication table of a number, display the square of each multiple, and count the total multiples generated.
6. Write a Python program to calculate the factorial of a number, count its digits, and display the reverse of the number.
7. Write a Python program to generate the Fibonacci series, calculate its sum, and display the largest term generated.
8. Write a Python program to create a list of numbers, find the largest element, and calculate the sum of all elements.
9. Write a Python program to remove duplicate elements from a list, sort the list, and display its length.
10. Write a Python program to count the occurrences of an element in a list, find its first position, and replace it with another value.
11. Write a Python program to reverse a string, count the vowels, and count the consonants.
12. Write a Python program to check whether a string is a palindrome, count the number of words, and convert it to uppercase.
13. Write a Python program to create a dictionary, add a new key-value pair, and display all keys and values separately.
14. Write a Python program to perform union, intersection, and difference operations on two sets.
15. Write a Python program to create a tuple, count the occurrences of an element, and find its index position.
16. Write a Python program to create a nested list, access its elements, and calculate the sum of all numeric values.
17. Write a Python program to accept marks of five subjects, calculate the total, percentage, and average.

18. Write a Python program to create a user-defined module, import the module, and call its functions.
19. Write a Python program to use the `math` module to calculate square root, power, and factorial.
20. Write a Python program to use the `random` module to generate random integers, random floating-point numbers, and random choices from a list.
21. Write a Python program to use the `datetime` module to display the current date, current time, and the day of the week.
22. Write a Python program to use the `os` module to display the current working directory, list all files, and create a new directory.
23. Write a Python program to demonstrate list slicing, tuple slicing, and string slicing.
24. Write a Python program to create a matrix using nested lists, display the matrix, and calculate row-wise sums.
25. Write a Python program to accept N numbers from the user, store them in a list, and display the maximum, minimum, and average values.

Assignment 2: Functions, File Handling, and Exception Handling

1. Write a Python program to create a user-defined function, call the function, and display the returned value.
2. Write a Python program to find the factorial of a number using a function, display the result, and validate the input.
3. Write a Python program to check whether a number is prime using a function, count its factors, and display the result.
4. Write a Python program to find the largest of three numbers using a function, calculate their average, and display the smallest number.
5. Write a Python program to demonstrate default arguments, keyword arguments, and positional arguments.
6. Write a Python program to demonstrate the use of `*args`, calculate the sum, and find the maximum value.
7. Write a Python program to demonstrate the use of `**kwargs`, display all key-value pairs, and count the arguments passed.
8. Write a Python program to calculate the factorial using recursion, count recursive calls, and display the result.
9. Write a Python program to calculate the sum of digits using recursion, reverse the number, and count its digits.
10. Write a Python program using a lambda function to find the square, cube, and double of a number.
11. Write a Python program using `map ( )` to calculate squares, cubes, and square roots of list elements.
12. Write a Python program using `filter ( )` to extract even numbers, odd numbers, and numbers greater than a specified value.
13. Write a Python program using `reduce ( )` to calculate the sum, product, and maximum value of list elements.
14. Write a Python program to create a text file, write data into it, and close the file properly.
15. Write a Python program to read a text file, count the number of lines, and display the contents.
16. Write a Python program to append data to a file, display the updated contents, and count the total words.
17. Write a Python program to copy the contents of one file to another, compare both files, and display the file sizes.
18. Write a Python program to search for a word in a file, count its occurrences, and replace it with another word.

19. Write a Python program to create multiple files, list all created files, and delete a selected file.
20. Write a Python program to handle division-by-zero exceptions, display a custom error message, and continue execution.
21. Write a Python program to handle multiple exceptions, display the exception type, and log the error message.
22. Write a Python program to demonstrate `try`, `except`, and `finally`, and display execution flow messages.
23. Write a Python program to raise a custom exception, handle the exception, and display a user-friendly message.
24. Write a Python program to handle a file-not-found exception, create the missing file, and write default content into it.
25. Write a Python program to create a custom exception class, validate user input, and handle invalid data.

Assignment 3: Decision Making and Conditional Statements

1. Write a Python program to accept a number, check whether it is positive, negative, or zero, and display an appropriate message.
2. Write a Python program to accept a number, check whether it is even or odd, and display its square.
3. Write a Python program to accept two numbers, find the larger number, and check whether both numbers are equal.
4. Write a Python program to accept three numbers, find the largest number, smallest number, and their average.
5. Write a Python program to accept marks of five subjects, calculate total marks, percentage, and grade using `if-elif-else`.
6. Write a Python program to check whether a student has passed or failed based on marks in each subject, total marks, and percentage.
7. Write a Python program to check whether a year is a leap year, century year, or ordinary year.
8. Write a Python program to check whether a character is a vowel, consonant, digit, or special symbol.
9. Write a Python program to check whether a character is uppercase, lowercase, digit, or special character.
10. Write a Python program to check whether a number is divisible by 5, 11, both 5 and 11, or neither.
11. Write a Python program to check whether a number is divisible by 2, 3, and 6 using logical operators.
12. Write a Python program to accept the age of a person, check voting eligibility, senior citizen status, and driving eligibility.
13. Write a Python program to calculate electricity bill charges based on unit slab rates and display the final bill amount.
14. Write a Python program to calculate income tax based on income slabs and display the tax amount.
15. Write a Python program to create a menu-driven calculator using `if-elif-else` for addition, subtraction, multiplication, and division.
16. Write a Python program to create a menu-driven program to find area of circle, rectangle, triangle, and square.
17. Write a Python program to check whether three sides form a valid triangle, identify its type, and calculate its perimeter.

18. Write a Python program to check whether three angles form a valid triangle and identify whether it is acute, right, or obtuse.
19. Write a Python program to accept month number, display the month name, and show the number of days in that month.
20. Write a Python program to accept a weekday number, display the weekday name, and identify whether it is a weekday or weekend.
21. Write a Python program to accept temperature, classify it as cold, normal, hot, or very hot, and convert it between Celsius and Fahrenheit.
22. Write a Python program to accept a password, check its length, validate whether it contains letters and digits, and display password strength.
23. Write a Python program to accept a product price, apply discount based on price range, and calculate the final price.
24. Write a Python program to accept employee salary, calculate bonus based on salary range, and display total salary.
25. Write a Python program to accept a number and use a conditional expression to find whether it is even or odd, positive or negative, and greater than 100.

Assignment 4: Object-Oriented Programming (OOP)

1. Write a Python program to create a class, create multiple objects, and display object details.
2. Write a Python program to use a constructor, initialize object attributes, and display object information.
3. Write a Python program to demonstrate instance variables, modify their values, and display the updated values.
4. Write a Python program to demonstrate class variables, modify the class variable, and observe changes across objects.
5. Write a Python program to demonstrate instance methods, call methods using objects, and display results.
6. Write a Python program to demonstrate class methods, static methods, and instance methods in the same class.
7. Write a Python program to implement single inheritance, access parent members, and override a method.
8. Write a Python program to implement multilevel inheritance, access inherited members, and display the inheritance hierarchy.
9. Write a Python program to implement multiple inheritance, access members from multiple parent classes, and resolve method calls.
10. Write a Python program to implement hierarchical inheritance, create objects of child classes, and access common parent methods.
11. Write a Python program to demonstrate method overriding, call overridden methods, and compare outputs.
12. Write a Python program to demonstrate encapsulation using private variables, getter methods, and setter methods.
13. Write a Python program to implement polymorphism using method overriding, create multiple objects, and invoke the same method.
14. Write a Python program to create an abstract class, implement abstract methods, and instantiate child classes.
15. Write a Python program to demonstrate abstraction using the `abc` module, implement multiple child classes, and display outputs.
16. Write a Python program to overload arithmetic operators, compare objects, and display the results.
17. Write a Python program to override the `__str__()` method, print object details, and compare with default output.
18. Write a Python program to demonstrate object composition, access composed objects, and display their details.

19. Write a Python program to create a package, import modules from the package, and call their functions.
20. Write a Python program to use the `collections` module, create a Counter, and display frequency analysis.
21. Write a Python program to use the `itertools` module, generate permutations, combinations, and Cartesian products.
22. Write a Python program to use the `calendar` module, display a monthly calendar, yearly calendar, and weekday information.
23. Write a Python program to design a Student class, store student details, and calculate grades.
24. Write a Python program to design an Employee class, calculate salary, and generate employee reports.
25. Write a Python program to build a Student Management System using OOP concepts, file handling, and exception handling.
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